

Project Proposal Preview

Request: add a search bar to my website

1. Requirements

Add a search bar to the plain HTML/CSS/JS website

Functional requirements

- Provide an input field with placeholder text and a submit button
- Capture user query and trigger search on submit or enter key
- Display matching results in a dropdown or results section
- Highlight matched terms in results
- Allow navigation to result links
- Ensure responsive layout for mobile and desktop

Non-functional requirements

- Search operation completes within 200ms for typical queries
- Compatible with latest versions of Chrome, Firefox, Safari, Edge
- Accessible: keyboard navigation, ARIA labels, sufficient contrast
- Maintainable: modular JS, clear comments
- Secure: escape user input to prevent XSS
- Minimal impact on page load time

Edge cases

- Empty query triggers no results or shows all items
- Queries with special characters handled gracefully
- No matching results displays a friendly message
- Very long queries truncated or limited
- Network latency simulated for async search
- Duplicate results deduplicated

Assumptions

- Site has static content that can be searched client-side or via a simple API endpoint
- No user authentication or personalized search required
- Search index is pre-generated or built on page load
- Browser supports ES6
- Search results are limited to visible page elements
- No pagination needed for result sets

2. Architecture

Add a lightweight client-side search module using vanilla JavaScript that loads a static JSON index and renders results dynamically.

Component	Responsibility	Type
SearchUI	Render input, button, results dropdown and handle user interactions	frontend

Component	Responsibility	Type
SearchEngine	Load index, filter results, highlight matches, deduplicate	frontend
SearchIndex	Static JSON file containing searchable content	integration

Data flow

- Page loads and SearchEngine fetches search-index.json asynchronously
- User focuses input and types query
- On keyup or submit, SearchEngine filters index based on query
- Filtered results are passed to SearchUI which renders a dropdown list
- Matches are highlighted and each item links to its target URL
- User selects a result, browser navigates to the linked page

File structure

- index.html
- assets/css/search.css
- assets/js/search.js
- assets/data/search-index.json

3. Tools

Tool	Category	Purpose	Required
Fuse.js	frontend	Client-side fuzzy search engine for filtering index	Yes
Mark.js	frontend	Highlight matched terms in search results	Yes
ESLint	frontend	Linting for maintainable, secure JavaScript code	Yes
GitHub Pages	hosting	Free hosting for static HTML/CSS/JS site	Yes

4. Cost Estimate (tools/services only)

Tool	Pricing model	Est. cost	Notes
Fuse.js	free	\$0	Open source library, no cost
Mark.js	free	\$0	Open source library, no cost
ESLint	free	\$0	Open source linting tool, no cost
GitHub Pages	free	\$0	Free hosting for static sites

Estimated monthly total: \$0

Note: These are third-party tool/service cost estimates only, not a cost of building the feature